Lights: strobe and beacon timing, interior lighting presets

The wing strobes flash the double flash of the real Airbus LED strobes, the A380X beacons flash halfway between the strobes, and the flyPad interior lighting presets save and load again

Aircraft	A32NX and A380X
Where to find it	Exterior lights: overhead EXT LT panel, STROBE and BEACON switches, seen from the outside view; interior lighting presets: flyPad > Presets
Branch	`core/lights`, based on `master` (FlyByWire master 66021c244 merged in); independent.
Status	Strobe and beacon timing checked in the sim by the user at night on both aircraft: OK. Lighting presets tested in the sim on the A32NX (save, load of a saved row, load of an unsaved row at 50 %): OK; the A380X uses the same fixed code but was not sim-tested. A32NX dome light with the DOME switch: native unit test, sim test pending.
Pictures	No pictures: a flash timing does not show on a still picture, and the presets page is the stock FBW flyPad page.

1. Strobe double flash and beacon phase

> **Aircraft:** A32NX and A380X | **Where:** EXT LT panel: STROBE ON, BEACON ON; outside view | **Status:** Checked in the sim by the user at night on both aircraft ("both look right").

What it is for

The FCOM gives no flash rates (A320 DSC-33-20-20, A380 DSC-33-30-20); the A380 FCOM only says that the strobes and the beacons alternate. The timing of the real lights comes from the Collins LED light data: a wing strobe double flash of 50 ms on, 60 ms off, 50 ms on, about 60 times a minute, the beacons flashing between the strobes. The FBW effects had a 50 ms gap (A32NX) and a 150 ms gap (A380X) in the double flash, and the A380X beacons flashed too close to the strobes.

What changed

Light	Before	Now
A32NX wing strobes	double flash 50-50-50 ms	double flash **50-60-50 ms**, 60 per minute (unchanged rate)
A380X wing strobes	double flash 50-150-50 ms	double flash **50-60-50 ms**, 45 per minute (unchanged rate)
A380X beacons	at +0.5 s in a 1.33 s cycle	at **+0.67 s**: exactly halfway between two strobe flashes
A32NX beacon, tail strobe	-	unchanged (beacon already halfway, at +0.5 s in a 1 s cycle)

Limitations and why

Limitation	Why	What to do
The A380X keeps its 45 flashes per minute.	There is no A380 source for the rate; 45 per minute is inside the Collins LED range (45 +/- 2).	-
A full MSFS restart is needed after an update.	The effect files (.fx) are loaded at sim start.	Restart the sim after installing.

How to use

1. Nothing to set: switch the **STROBE** and **BEACON** lights ON on the overhead EXT LT panel as usual.

How to test

#	Do this	Expected result
1	A32NX at night (or dusk), on the ground. Switch **STROBE ON** and **BEACON ON**. Go to an outside view that shows a wing tip and the fuselage.	Each wing strobe gives a crisp double flash (two short flashes very close together) about once a second.
2	Watch the beacon and the strobes together.	The beacon flashes halfway between two strobe double flashes: strobe, beacon, strobe, beacon, evenly spaced.
3	Repeat with the A380X.	Same crisp double flash of the wing strobes, a little slower (45 per minute); the beacons (top and belly) flash exactly halfway between the strobes, not right after them.

2. Interior lighting presets save and load again

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Presets | **Status:** Tested in the sim by the user on the A32NX: save, load of a saved row, load of an unsaved row at 50 %: OK. Not sim-tested: the load with no presets file yet (fresh install), and the A380X (same code). Native unit tests: extra-backend test/run_tests.sh.

What it is for

The flyPad interior lighting presets did nothing or the wrong thing on both aircraft, because of three bugs in the C++ extra-backend:

- **No preset was ever saved.** The shared ini library refused the existing work folder (it read the mkdir result the wrong way), so every save failed silently and no presets file was written. It now accepts an existing folder. The FADEC fuel configuration save uses the same library and benefits too.
- **An unsaved row faded every light to off.** Loading a row that was never saved left the target values unset, so every light, the PFD, ND, ECAM and MCDU brightness included, faded to 0. Such a row now loads the default preset, **all lights at 50 %**, as the flyPad page says. The A32NX default also had its DCDU and MCDU values in the wrong scale (0.5 instead of 50).
- **A load before the first save failed.** With no presets file yet (fresh install), a load now gives the default 50 % preset instead of leaving the lights untouched.
- **A32NX: the dome light came on without its switch.** The captain flyPad auto-loads a preset at every power-up, and the preset code set the dome light to 0, 50 or 100 %, while the overhead DOME switch only has OFF (0), DIM (20) and BRT (100): the lamps lit, the switch stayed at OFF, and a preset saved at DIM came back at 50 %. A preset now moves the dome light to the nearest switch position, so the switch follows, and the default preset keeps the dome light **off**.

What you should see

Situation	What you should see
Save Preset on a row (aircraft powered)	The current lighting levels are stored; the row loads them later, also after a sim restart.
Load Preset on a saved row	The cockpit lights and display brightness fade to the saved levels.
Load Preset on a row never saved	Every light and display brightness fades to 50 %, the A32NX dome light stays off; nothing goes dark.
Automatic loading (Day / Dawn/Dusk / Night set to a row)	The selected row loads as above: its saved levels, or the 50 % default if it was never saved. On the A32NX the dome light only comes on when the saved preset had it on.

A32NX DOME switch after a preset load	The switch sits at OFF, DIM or BRT, matching the lamps.

Limitations and why

Limitation	Why	What to do
Presets saved before the fix do not exist.	No save ever reached the file before.	Save the rows again once.
The aircraft must be powered to save or load.	Stock flyPad rule.	Power the aircraft first.
A full MSFS restart is needed after an update.	The extra-backend WASM module changes.	Restart the sim after installing.

How to use

1. Power the aircraft and set the cockpit lights and display brightness as you like.
2. Open flyPad > Presets, optionally rename a row, and press the **Save** icon button on it.
3. Later, press **Load** on that slot to get the same lighting back.

How to test

#	Do this	Expected result
1	A32NX (or A380X) powered, on the ground. Set the panel and flood lights and the display brightness to clearly different levels. flyPad > Presets: press the **Save** icon button on slot 1 and confirm.	The save completes.
2	Change the lights, then press **Load** on slot 1.	The lights and displays fade back to the saved levels.
3	Press **Load** on a slot that was never saved.	Every light and display brightness fades to 50 %: the PFD, ND, ECAM and MCDU stay readable; on the A32NX the dome light stays off.
4	A32NX at night: DOME switch at DIM, save slot 2; switch to OFF, load slot 2.	The dome lamps come on dim and the switch moves to DIM.
5	Restart the sim, power the aircraft and load row 1 again.	The saved levels come back (the presets file was written).

3. Aircraft presets load reliably

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Presets (aircraft presets) | **Status:** Not sim-tested. The preset runner has a native unit test.

What it is for

The aircraft presets (Ready for pushback, taxi, take-off and the like) could run steps out of order or hang. They now run through a state machine: the expedited loading mode is latched when the load starts, so flipping the flyPad switch halfway no longer changes which steps run; after each action the runner waits (at most 2 s) for the expected state before the next step, without repeating toggles; a condition step that never becomes true aborts the preset after 180 s instead of hanging; the quick mode variable is cleared when the aircraft stops being ready in the middle of a load; and the procedure file loader no longer stops the module on a missing Name or Type attribute.

How to use

1. Open flyPad > Presets, choose the aircraft preset and press **Load**.

How to test

#	Do this	Expected result
1	Aircraft cold and dark. flyPad > Presets: load the aircraft preset for the first stage, with expedited loading on.	The steps run one after the other and the preset completes.
2	Start another load and flip the expedited loading switch halfway.	The load keeps the mode it started with.